Response to the third referee report

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The revision improves the free-angle reflection construction from six to five unrestricted two-qubit gates and adds exact native-count certificates to the atlas. The seven-CNOT reflection identity and the interior lower bounds remain open; we now state the five/seven claim as a conjecture. Earlier responses are preserved separately.

1. Trace-± 2 reflection family

For $R_\theta = I - 2|\psi_\theta\rangle\langle\psi_\theta|$, with $|\psi_\theta\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle$, we give an explicit unitary B_θ satisfying

$$B_\theta R_\theta X_a B_\theta^\dagger = Z_a S_b.$$

The five gates in time order are

$$\mathrm{CX}_{c,a}, \quad B_{\theta,ab}, \quad CS_{c,b}, \quad CZ_{c,a}, \quad B_{\theta,ab}^\dagger.$$

The zero-control branch is identity and the one-control branch is $B_\theta^\dagger Z_a S_b B_\theta X_a = R_\theta$. The manuscript supplies the matrix and basis convention. Exact symbolic multiplication verifies unitarity and the full literal identity with θ free. The sign change adds a local control Z. The proved interior bounds are consequently

$$4 \le E_{\mathrm{CX}} \le 8, \qquad 3 \le E_{\mathrm{arb}} \le 5.$$

We independently found a seven-CNOT layout with BQSKit and refined its local gates at ten angle values, including $\pi/8$ and 0.3. The archived words use seven CXs and physical U3 locals, with no GP gate. Independent Qiskit matrix replays have maximum literal entry error below 10^{-12}. These are numerical witnesses, not a free-angle upper-bound proof. The seeds, layout, complete words, versions, replay report, and reproduction harness are included.

We also prove a partial doubled-spectrum obstruction for $R_\theta(A\otimes I)$, for any one-qubit unitary A; the supplement gives the determinant argument and its limited layout scope. We have not extended Yu–Ying's exclusion to every four-gate layout. Their Fredkin reductions use literal wire-SWAP identities that do not automatically apply to the interior reflections. The main paper therefore states the conjecture $E_{\mathrm{arb}} = 5$ and $E_{\mathrm{CX}} = 7$ for $0 < \theta \le \pi/4$ explicitly, without claiming that numerical failures prove a lower bound.

2. Monodromy-polytope certificates for native columns

We implemented exact count certificates from the corrected Peterson–Crooks–Smith inequalities. The quantum Littlewood–Richardson data retain their authors' source revision, hash, attribution,

and license. The implementation includes both projective-alcove branches at every composition. Numerical eigenangles and linear programming propose candidates only: exact complete Makhlin invariants accept target coordinates, rational primal points prove reachability, and rational Farkas multipliers exclude all branches at smaller depths.

The original numerical audit contained 150 target/native pairs in the SYC and square-root-SWAP columns. The first exact pass proves 111 minimum counts and one lower bound. A deeper query resolves that last record: SYC needs exactly five square-root-SWAP pulses with arbitrary local gates; the displayed atlas word still uses six. All 64 fixed-target/native pairs in this subset now have exact minimum-count certificates. The other proved records concern named parameter substitutions. Thirty proof deadlines and eight uncertified coordinate conversions remain explicitly inconclusive.

Examples include three square-root-SWAPs for CS and B, four for CT and iSWAP, and two SYCs for B. Thus the audit also certifies that three square-root-SWAPs are not universal. An independent rational checker reconstructs every accepted feasibility and exclusion certificate.

These results prove counts with arbitrary local $U(2)$ frames. The supplied U3 and PhasedXZ alphabets generate those frames without a GP gate through the checked phase-absorption identities. They do not give explicit shortest symbolic frame words or local-count minima. Atlas panels and the CSV now distinguish proved fixed-target minima, proved counts at one family sample, and remaining numerical leads. A hard sampled member plus a matching full-family construction can separately certify a minimum uniform family budget, which is labelled as such. Displayed long constructions retain their visible gaps; no circuit is silently replaced by an existence proof.

3. Supplement layout and editorial fields

The empty implementation-objective heading has been removed. The substantive objective and reproducibility material remains with its explanatory text. The main manuscript now has the author-supplied affiliations for Gregory Morse and Péter Rakyta. Gregory is credited with the research, software, mathematical analysis, validation, experiments, data, visualization, and original draft; Péter with scientific discussions, drafting, reviewing, and editing.

We also supply an unpublished-review version using the official `quantumarticle` class. Its wrapper inputs the same main manuscript; the class, license header, and provenance are included with the editable sources. No external submission has been performed.

4. Fredkin endpoint wording

The paper now says plainly that seven CNOTs is the best known construction count in our literature audit, not a proved minimum. The proved Fredkin unrestricted minimum is five; its displayed CNOT bounds remain five to seven. The interior conjecture is separate from both endpoint facts.